

Robust analog two-qubit gates from deep reinforcement learning: an independent reproduction and family-resolved assessment

Francesco Giuseppe Minisini^{1,*}

¹*Dipartimento di Fisica “Aldo Pontremoli”, Università degli Studi di Milano, Via Celoria 16, 20133 Milano, Italy*
(Dated: September 7, 2026)

Reinforcement learning has been proposed as a unified route to analog quantum gate design, in which accuracy, leakage suppression, runtime and robustness against stochastic control errors are optimized jointly rather than sequentially. We report an independent, from-scratch reimplementation and critical assessment of the Universal cost Function control Optimization (UFO) framework of Niu *et al.* [npj Quantum Inf. **5**, 33 (2019)] for the two-qubit gate family $N(\alpha, \alpha, \gamma) = \exp[i(\alpha\sigma^x\sigma^x + \alpha\sigma^y\sigma^y + \gamma\sigma^z\sigma^z)]$ on a three-level-per-mode gmon model, including the second-order time-dependent Schrieffer–Wolff leakage bound, the exact bandwidth filter, and trust-region policy optimization in a stochastic control environment. Against a deliberately strong direct gradient (GRAPE-style, Adam) baseline optimized on the same Hamiltonian and the same objective, we find that the qualitative claims of the original work reproduce, while several quantitative ones do not. On a fixed nominal target the direct optimizer *outperforms* the policy-gradient controller in cost, runtime and leakage, so the value of the learning formulation is not nominal optimality. Stochastic training is nonetheless unambiguously effective as a robustness mechanism: against an otherwise identical agent trained deterministically, the noise-trained policy attains higher average fidelity and lower fidelity variance at *every* noise strength in $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz. Its advantage over the noise-aware direct optimizer, however, is confined to a neighbourhood of the training noise: the two curves cross at $\sigma_{\text{noise}} \approx 1.3$ MHz, beyond which the direct optimizer is both more accurate and more stable. Learned robustness is therefore local to the perturbation scale seen during training rather than a global property, and the margins we measure ($1.4\times$ in infidelity over deterministic RL at 1 MHz) lie well below the one-to-two orders of magnitude originally reported. A curriculum sweep across $N(\alpha, \alpha, \pi/2)$ shows that the analog runtime advantage over optimal gate synthesis is real but strongly structured, ranging from 16% at the edges of the family to a factor 2.8 in a broad central basin and collapsing to a 2 ns solution at the hardware-aligned point $\alpha = \pi/2$; sweeping the same family with the direct optimizer alone, however, yields shorter pulses still, so the speedup belongs to analog pulse-level control rather than to the learning method. We release the full parameter set needed to reproduce these numbers.

I. INTRODUCTION

A quantum algorithm is written as a sequence of logical unitaries, but a quantum processor executes continuous-time dynamics generated by an experimentally accessible Hamiltonian. On pre-fault-tolerant superconducting hardware the translation between the two is not a formality: coherence budgets, calibration drift, waveform distortion and multilevel structure all limit which circuits can actually be run [1–3]. Quantum control is therefore not a fine-tuning layer on top of compilation, but one of the binding constraints on usable computational depth.

This observation motivates a shift of perspective. Rather than decomposing a target two-qubit unitary into a fixed universal gate alphabet and paying the runtime cost of every elementary operation, one may attempt to realize it directly through a single shaped analog pulse that exploits the native interactions of the device [4, 5]. The appeal is concrete: if the target is well aligned with the hardware’s own entangling generator, the analog implementation can be substantially shorter than any compiled circuit. The difficulty is equally concrete. Super-

conducting modes are weakly anharmonic, so fast or aggressively modulated pulses drive population out of the computational subspace [6, 7]; and the pulse that reaches the device is never the pulse that was designed, because amplitudes fluctuate and Hamiltonian parameters are known only approximately.

Niu *et al.* [4] proposed to attack these requirements simultaneously rather than in sequence. Their Universal cost Function control Optimization (UFO) framework combines three ingredients: an analytic leakage estimate derived from a time-dependent Schrieffer–Wolff transformation (TSWT), a scalar objective that adds leakage, boundary and runtime penalties to gate infidelity, and a continuous-action reinforcement-learning (RL) formulation trained with trust-region policy optimization (TRPO) [8] in an environment where control noise is injected during training. The reported outcome was strong: up to one order of magnitude lower average gate infidelity than an RL agent trained without noise, roughly two orders of magnitude lower than a stochastic gradient-descent baseline, about one order of magnitude smaller fidelity variance, and a factor-ten runtime reduction relative to optimal gate synthesis on the subfamily $N(\alpha, \alpha, \pi/2)$.

These are consequential claims, and they have been influential in establishing RL as a credible language for

* fg.minisini@gmail.com

robust pulse design [9–11]. They have not, to our knowledge, been subjected to an independent reimplementa- tion. That is the purpose of this work.

What this paper does. We rebuild the framework from scratch — multilevel gmon simulator, second-order TSWT leakage estimator, exact bandwidth filter, UFO objective, TRPO agent and stochastic environment — and use it to ask four questions. (i) Does the frame- work reproduce under an independent implementation, and with what quantitative fidelity? (ii) How does the policy-gradient controller compare against a *strong* direct pulse optimizer, rather than a weak one, when both are given the same Hamiltonian, the same constraints and the same objective? (iii) Is the improvement under noise attributable to reinforcement learning as such, or specifically to stochastic training? (iv) How robust are the conclusions to the free parameters that the original paper does not pin down, namely the actor-critic archi- tecture and the UFO weights?

What we find. The qualitative picture survives; sev- eral numbers do not. Our central results are:

1. On the representative nominal target $N(2.2, 2.2, \pi/2)$, a GRAPE-style Adam opti- mizer beats the nominal TRPO controller on *every* nominal metric — total cost (0.158 vs 0.298), runtime (70 ns vs 116 ns) and leakage (2.8×10^{-4} vs 1.6×10^{-3}). Policy-gradient control is not intrinsically a better nominal pulse optimizer.
2. Stochastic training works, and the controlled com- parison is unambiguous: the noise-trained agent has higher average fidelity and lower fidelity vari- ance than the deterministically trained agent at *every one* of the 3401 noise strengths tested. Because the two share architecture, objective and algorithm, this isolates stochastic *training* as the operative in- gredient.
3. That advantage is nonetheless local to the training scale, and smaller than originally reported. Rela- tive to the nominal agent the infidelity ratio falls from 3.8 at $\sigma_{\text{noise}} = 0.1$ MHz to 1.44 at 1 MHz. Against the noise-aware Adam baseline the noise- trained policy wins only below $\sigma_{\text{noise}} \approx 1.3$ MHz; beyond that crossover the direct optimizer is both more accurate and more stable, and it is the slowest-degrading controller overall.
4. The runtime advantage over optimal gate synthe- sis is real across the entire family $N(\alpha, \alpha, \pi/2)$, but ranges from 16% at the edges to $2.8\times$ in the cen- tral basin, with a singular hardware-aligned point at $\alpha = \pi/2$ solved in 2 ns. The advantage tracks alignment with the native exchange interaction, ex- actly as the physical argument of Ref. [4] predicts.
5. It is not, however, a benefit of the learning method. Sweeping the same family with the direct optimizer

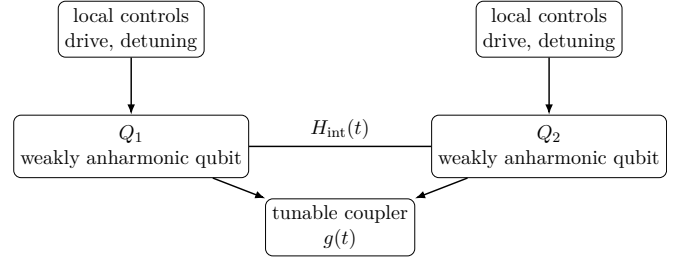


FIG. 1. The gmon control setting. Two weakly anharmonic modes are driven locally and coupled through a tunable interaction $g(t)$. Local addressability together with a time-dependent coupling gives the seven control channels of Eq. (2).

alone gives shorter pulses than the curriculum-trained policy at every target but $\alpha = \pi/2$ (median $3.1\times$ over synthesis, against $1.5\text{--}2.0\times$), at compara- ble fidelity and robustness. The speedup is a prop- erty of analog pulse-level control, not of reinforc- ement learning.

Section II states the framework; Sec. III gives the im- plementation and the complete parameter set; Sec. IV fixes the free parameters through two sweeps; Secs. V–VII present the nominal, robustness and runtime results; Sec. VIII confronts them with the original claims.

II. CONTROL FRAMEWORK

A. Physical model and control channels

The platform is the gmon architecture, sketched in Fig. 1: two weakly anharmonic superconducting modes with local drive and frequency control, coupled through a flux-tunable interaction [5]. The tunability of the cou- pling is what makes the platform interesting for analog gate design, because the entangling interaction becomes a control resource rather than a fixed background.

After circuit quantization, transformation to the inter- action picture and a rotating-wave approximation, the effective two-mode Hamiltonian is [4]

$$\begin{aligned}
 \hat{H}_{\text{RWA}}(t) = & \frac{\eta}{2} \sum_{j=1}^2 \hat{n}_j (\hat{n}_j - 1) + g(t) (\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2) \\
 & + \sum_{j=1}^2 \delta_j(t) \hat{n}_j \\
 & + \sum_{j=1}^2 i f_j(t) \left(\hat{a}_j e^{i\phi_j} - \hat{a}_j^\dagger e^{-i\phi_j} \right), \quad (1)
 \end{aligned}$$

with $\hat{n}_j = \hat{a}_j^\dagger \hat{a}_j$. The four terms carry distinct physical roles. The anharmonicity η breaks the harmonic ladder and makes the lowest two levels of each mode a usable qubit; because η is finite, the system stays intrinsically multilevel and leakage remains a real error channel. The

exchange term is the entangling resource. The detunings act diagonally and shape relative phase accumulation. The drives generate local rotations. The available controls are therefore

$$\mathbf{u}(t) = \{g(t), \delta_1(t), \delta_2(t), f_1(t), f_2(t), \phi_1(t), \phi_2(t)\}. \quad (2)$$

Projected onto the logical subspace, Eq. (1) becomes

$$\begin{aligned} \hat{H}_{\text{qb}}(t) = & \frac{g(t)}{2} (\sigma_1^x \sigma_2^x + \sigma_1^y \sigma_2^y) \\ & + \sum_j \left[\frac{\delta_j}{2} \sigma_j^z - f_j (\sin \phi_j \sigma_j^x + \cos \phi_j \sigma_j^y) \right], \end{aligned} \quad (3)$$

which shows that the coupling generates an XY -type entangler while the local channels supply arbitrary single-qubit rotations — enough for universal two-qubit control. Equation (3) is an interpretive tool only; all dynamics in this work are propagated with the full multilevel Eq. (1), because the whole point is that leakage must not be projected away.

B. Subspaces and the TSWT leakage bound

In the Fock basis $|n_1 n_2\rangle$ the computational sector is $\Omega_0 = \text{span}\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ and the leading leakage sector is $\Omega_1 = \text{span}\{|02\rangle, |20\rangle, |12\rangle, |21\rangle\}$. Writing the physical space as $\mathcal{H} = \bigoplus_{\alpha} \Omega_{\alpha}$ and decomposing

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) + \hat{H}_2(t), \quad (4)$$

the diagonal drift $\hat{H}_0$ sets the subspace energies, $\hat{H}_1$ is block-diagonal (evolution within each sector) and $\hat{H}_2$ is block-off-diagonal (direct leakage). With Δ the minimum gap between Ω_0 and the leakage sectors and $\epsilon \sim \|\hat{H}_1\| \sim \|\hat{H}_2\|$ the control scale, the regime of interest is $\epsilon/\Delta \ll 1$.

The essential difficulty is that $\hat{H}_2$ cannot simply be switched off: the same controls that produce the desired logical rotations also generate the inter-subspace coupling. Leakage suppression is therefore a control problem, not a projection argument. The TSWT addresses it by moving to a rotated frame $|\tilde{\psi}\rangle = e^{-\hat{S}(t)} |\psi\rangle$ with anti-Hermitian $\hat{S}$, in which

$$\hat{\mathbb{H}}(t) = -i \frac{d e^{-\hat{S}}}{dt} e^{\hat{S}} + e^{-\hat{S}} \hat{H} e^{\hat{S}}. \quad (5)$$

The first term is the price of a moving basis: in a driven system, leakage is generated not only by direct coupling but also by the non-adiabatic motion of the frame itself. Expanding $\hat{S} = \epsilon \hat{S}_1 + \epsilon^2 \hat{S}_2 + \dots$ and cancelling block-off-diagonal terms order by order leaves, at second order, a residual $\hat{H}_{\text{od}} = O(\epsilon^3/\Delta^2)$, from which the accumulated coherent leakage over a gate of duration T is bounded by

$$\begin{aligned} L_{\text{tot}}(T) \lesssim & \frac{\|\hat{H}_{\text{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\text{od}}(T)\|}{\Delta(T)} \\ & + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\text{od}}(t)}{dt^2} \right\| dt. \end{aligned} \quad (6)$$

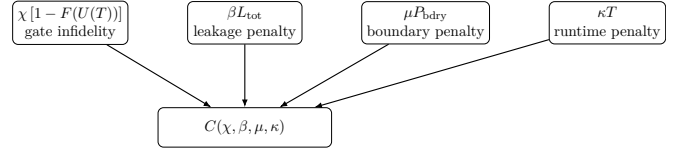


FIG. 2. The UFO objective, Eq. (8), scalarizes four competing physical requirements: logical accuracy, accumulated leakage, boundary regularity and total runtime.

A derivation is summarized in Appendix A. The structure of Eq. (6) is what makes it useful for optimization rather than merely for diagnosis. The two boundary terms penalize residual off-diagonal coupling at the pulse endpoints, and therefore push the controls toward configurations where the dressed and bare bases coincide at $t = 0$ and $t = T$. The integral term penalizes rapid modulation of $\hat{H}_{\text{od}}$ and is the quantitative expression of the speed-leakage trade-off: a pulse can be small in amplitude at every instant and still leak, if it changes too quickly. Crucially, Eq. (6) is written in terms of the Hamiltonian and its derivatives, so it can be accumulated online during propagation instead of requiring a separate population measurement at the end of the pulse.

C. The UFO objective

Let $U(T) = \mathcal{T} \exp[-i \int_0^T \hat{H}_{\text{RWA}}(t) dt]$ and let U_{proj} be its restriction to Ω_0 . The gate fidelity with respect to U_{target} is

$$F = \frac{1}{16} |\text{Tr}(U_{\text{proj}}^\dagger(T) U_{\text{target}})|^2, \quad (7)$$

which equals 1 exactly when the implemented gate matches the target up to a global phase. The full objective adds three penalties to this term, as summarized in Fig. 2 [4]:

$$\begin{aligned} C(\chi, \beta, \mu, \kappa) = & \chi[1 - F] + \beta L_{\text{tot}} + \kappa T \\ & + \mu \sum_{t \in \{0, T\}} [g(t)^2 + f_1(t)^2 + f_2(t)^2]. \end{aligned} \quad (8)$$

The boundary term enforces vanishing envelopes at the pulse edges, which both permits clean gate concatenation and reduces errors from deviations of the rotating-wave approximation; the linear runtime penalty encodes the fact that long gates buy more exposure to decoherence and drift.

The design choice that matters here is that all of these are *soft* constraints. Much earlier leakage-mitigation work proceeded by disabling control channels or imposing structural restrictions on the admissible pulse shape, which suppresses leakage at the price of controllability. UFO keeps every channel available and penalizes undesirable behaviour instead, leaving the optimizer free to trade the four objectives against one another. This also

means the framework is modular: on a different platform the individual terms can be replaced without changing the logic.

D. Reinforcement-learning formulation

The pulse is built sequentially. Time is discretized in steps Δt ; the state presented to the agent is a representation of the propagator accumulated so far, $s_n \sim U_n$; the action is the control vector for the next interval, $a_n = (g_n, \delta_{1,n}, \delta_{2,n}, f_{1,n}, f_{2,n}, \phi_{1,n}, \phi_{2,n})$; and the environment propagates

$$U_{n+1} = \exp[-i\Delta t(\hat{H}_{n+1} + \delta\hat{H}_{n+1})]U_n, \quad (9)$$

where $\delta\hat{H}$ is the stochastic perturbation used in noisy training. The reward is the negative running UFO cost, $r_n = -c_n$, so maximizing expected return minimizes the expected objective. An episode terminates at $\min(T_{\max}, T_{\text{stop}})$, where T_{stop} is the first time the cost falls below a threshold; the agent thus co-optimizes pulse shape *and* stopping time, which is a structural difference from fixed-horizon pulse optimization.

Using the accumulated unitary as the state is a deliberate choice: the agent reasons in terms of how much of the target operation has already been built, rather than in terms of local control variables, which matches the structure of gate synthesis.

Optimization uses TRPO [8], which maximizes the importance-weighted surrogate

$$\mathcal{L}(\theta) = \mathbb{E}_{\pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} \hat{A}(s, a) \right] \quad \text{s.t.} \quad \mathbb{E}_s [D_{\text{KL}}] \leq \delta_{\text{KL}}. \quad (10)$$

The trust-region constraint matters here for a specific reason: the UFO reward couples four competing terms and the environment is stochastic, so the landscape is irregular and a single over-large gradient step can destroy previously learned behaviour. Constraining the step in distribution space buys stability at the cost of speed.

Robustness enters through the environment rather than through an outer loop. In noise-optimized training, Gaussian fluctuations are added to the control amplitudes at every step, so the objective becomes an expectation over both the policy and the noise, $J(\theta) = -\mathbb{E}_{\tau, \xi}[C]$. Pulse trajectories that are fragile under small perturbations earn poor average return and are disfavoured; stochastic training therefore acts as an implicit regularizer. Throughout this paper we call *nominal* an agent trained in the deterministic environment and *noise-optimized* one trained in the stochastic environment. They differ only in the training distribution — same architecture, same objective, same algorithm — which is what makes their comparison informative.

E. Target family and synthesis reference

Benchmarking uses the family

$$N(\alpha, \alpha, \gamma) = \exp[i(\alpha\sigma_1^x\sigma_2^x + \alpha\sigma_1^y\sigma_2^y + \gamma\sigma_1^z\sigma_2^z)]. \quad (11)$$

The equal XX and YY coefficients mirror the exchange structure of the gmon coupling, so the family is deliberately close to the hardware's native interaction while remaining rich: up to single-qubit rotations it contains SWAP, iSWAP, CNOT and CZ, as well as the fermionic SWAP and Givens rotations that appear in Jordan–Wigner-mapped fermionic simulation [4]. Because it is continuously parametrized, one can study an entire manifold of related targets rather than isolated cases.

The runtime reference is the optimal circuit decomposition of $N(\alpha, \alpha, \gamma)$, which by the Vatan–Williams construction requires three entangling gates and a small number of single-qubit layers [12]. With the state-of-the-art gate times assumed in Ref. [4], $T_{1q} = 20$ ns and $T_{\text{CNOT}} = 45$ ns, this gives

$$T_{\text{syn}} \approx 215 \text{ ns}, \quad (12)$$

which we use unchanged as the compiled-implementation baseline.

III. NUMERICAL IMPLEMENTATION

A. Simulator

Each mode is truncated to three levels, giving $d_{\text{phys}} = 3^2 = 9$ and $d_{\text{comp}} = 4$. Hamiltonians are kept in rad ns^{-1} internally; one-step propagation uses exact diagonalization of the Hermitian generator rather than a series expansion. The base anharmonicity is $\eta = -200$ MHz, which also sets the gap scale entering Eq. (6). Control amplitudes are bounded at 20 MHz, well below $|\eta|$ so that the perturbative regime $\epsilon/\Delta \ll 1$ is respected, and phases are mapped to $[0, 2\pi]$.

Before reaching the Hamiltonian, the piecewise-constant controls pass through the two-pole double-exponential filter of Ref. [4], implemented exactly: with $a = \exp(-\pi B/f_s)$, $B = 10$ MHz the bandwidth and $f_s = 1/\Delta t$ the sampling rate,

$$\tilde{u}_k = (1 - a)^2 u_k + 2a \tilde{u}_{k-1} - a^2 \tilde{u}_{k-2}. \quad (13)$$

This is not cosmetic. It ensures that the optimizer cannot exploit arbitrarily sharp control transitions that no waveform generator could produce, and it interacts directly with the non-adiabatic term of the leakage bound.

The leakage estimator implements the second-order TSWT numerically: at each step it decomposes H_t into H_0, H_1, H_2 , solves for the generators S_1 and S_2 on the off-diagonal blocks, estimates $\dot{S}_1, \dot{S}_2$ by finite differences, forms the residual $H_{\text{od}}^{(3)} = [H_1, S_2] + \frac{1}{3}[[H_2, S_1], S_1] - i\dot{S}_2$, and accumulates the boundary and integral contributions

of Eq. (6) incrementally (Appendix B). The incremental form matters for cost: a naive implementation rescans the trajectory history at every step and is $O(T^2)$ per rollout.

B. Noise model

At each step the realized control is $\mathbf{u}^{(\text{noisy})}(t_k) = \mathbf{u}(t_k) + \boldsymbol{\xi}(t_k)$ with $\boldsymbol{\xi}$ drawn independently from a zero-mean Gaussian of standard deviation σ_{noise} . Perturbations are applied to the coupling g , the detunings $\delta_{1,2}$ and the drive amplitudes $f_{1,2}$, and additionally to the anharmonicity η ; phases are left noise-free. Including η is a deliberate feature of the model: it probes sensitivity not only to imperfect pulse delivery but also to uncertainty in a device parameter that the controller cannot actuate at all.

Training noise and evaluation noise are kept conceptually separate. The Adam baseline and the nominal TRPO agent are optimized without perturbations and probed *a posteriori*; the noise-optimized agent sees the same perturbation model during training, at $\sigma_{\text{train}} = 1$ MHz.

C. Controllers compared

Direct gradient baseline (Adam). This is intended as a strong comparator, not a foil. For a fixed target and a fixed horizon T , the full discretized trajectory $\{g_\ell, \delta_{j,\ell}, f_{j,\ell}, \phi_{j,\ell}\}_{\ell=0}^{N_T-1}$ is optimized directly with Adam [13] on the same multilevel model and the same objective Eq. (8) — a GRAPE-style direct pulse search [3, 14] rather than a literal reproduction of the canonical algorithm. Since the horizon is not a free variable of a single run, the procedure is repeated over $T \in \{40, 50, \dots, 180\}$ ns (400 Adam iterations each, learning rate 0.03) and the resulting plans are re-evaluated in a common deterministic simulator. The baseline is therefore a family of fixed-horizon problems, whereas TRPO learns a variable-time policy; this structural difference is precisely what makes the comparison informative.

Nominal and noise-optimized TRPO. Both use the same actor-critic pair. The observation vectorizes the current 9×9 propagator into real and imaginary parts and appends the normalized time index,

$$s_t = (\Re \text{vec}(U_t), \Im \text{vec}(U_t), t/T_{\text{max}}), \quad d_{\text{obs}} = 163. \quad (14)$$

Actor and critic are multilayer perceptrons with hidden sizes (64, 32, 32) and tanh activations, as in Ref. [4]. The actor emits a mean $\mu_\theta(s_t) \in \mathbb{R}^7$ with a state-independent trainable log σ ; raw actions $\tilde{a}_t = \mu_\theta + \sigma \odot \varepsilon_t$ are squashed by tanh into $[-1, 1]^7$ and rescaled to physical bounds. The critic outputs a scalar baseline for advantage estimation.

Gate synthesis. The compiled reference of Eq. (12), used for runtime only.

D. Metrics

Nominal quality is reported through the gate fidelity Eq. (7), the TSWT leakage estimate L_{tot} with its boundary/integral split, the boundary penalty, the effective runtime and the total cost Eq. (8).

Robustness is characterized by two distinct quantities, and the distinction is worth stating explicitly because they are often conflated. The first is the average fidelity of the *noise-averaged channel* $\mathcal{E}$, estimated from the Monte Carlo ensemble of noisy projected unitaries $\{K_j\}$ through the standard formula [15]

$$\bar{F} = \frac{\sum_j \text{Tr}(U_{\text{target}} U_j^\dagger U_{\text{target}}^\dagger \mathcal{E}(U_j)) + d^2}{d^2(d+1)}, \quad d = 4, \quad (15)$$

with $\{U_j\}$ a two-qubit Pauli basis. The second is the variance across realizations of the per-shot gate fidelity,

$$\text{Var}(F) = \mathbb{E}_\xi [(F_\xi - \mathbb{E}_\xi[F_\xi])^2]. \quad (16)$$

A controller with good mean fidelity but large variance is experimentally fragile, so neither quantity alone settles the question.

E. Parameter set

Table I collects every setting used. We report per-experiment values rather than a single idealized configuration, because the model-selection sweeps deliberately ran under a reduced budget and a looser stopping threshold than the production runs; conflating them would misstate what was actually done.

IV. FIXING THE FREE PARAMETERS

Two ingredients of the framework are not determined by the physics: the capacity of the actor-critic networks and the relative weights in Eq. (8). Both could silently drive the comparisons that follow, so we fix them first, on the representative target $N(2.2, 2.2, \pi/2)$, under a reduced budget and before any production run.

A. Actor-critic architecture

We swept five hidden-layer layouts of comparable scale, three seeds each, at 20 policy iterations and 5000 episodes per batch (Table II). Selection was not made on total cost alone, because a network can reach a favourable cost by trading leakage against fidelity or by settling on an unnecessarily long pulse; we therefore tracked nominal fidelity, leakage, runtime and robustness at $\sigma_{\text{noise}} = 1$ MHz simultaneously.

TABLE I. Complete configuration. Values in the last two columns are the settings actually used for the production runs and for the reduced-budget model-selection sweeps of Sec. IV.

Quantity	Symbol	Production	Sweeps
<i>Physical model</i>			
Levels per mode	—	3	3
Anharmonicity	η	−200 MHz	−200 MHz
Control bound	$u_{\max}$	20 MHz	20 MHz
Filter bandwidth	B	10 MHz	10 MHz
Timestep	Δt	2 ns	2 ns
Max gate duration	$T_{\max}$	180 ns	180 ns
Runtime norm.	T_{norm}	60 ns	60 ns
<i>UFO weights</i>			
Fidelity	χ	10	swept
Leakage	β	10	10
Boundary	μ	0.2	swept
Runtime	κ	0.1	swept
<i>TRPO</i>			
Hidden sizes	—	(64, 32, 32)	swept
Trust region	δ_{KL}	0.01	0.01
Discount	γ_{RL}	0.99	0.99
GAE parameter	λ	0.97	0.97
CG iters / damping	—	10 / 0.1	10 / 0.1
Initial exploration	$\log \sigma_0$	−0.5	−1.0
Value lr / epochs	—	10^{-3} / 50	10^{-3} / 50
Episodes / batch	—	5000 ^a	5000
Policy iterations	—	115 ^b	20
Stopping cost	C_{term}	0.30 ^c	0.40
Training noise	σ_{train}	1 MHz	1 MHz
<i>Adam baseline</i>			
Iterations per horizon	—	400	—
Learning rate	—	0.03	—
Horizon grid	—	40–180 ns, step 10 ns	—
<i>Robustness evaluation</i>			
Noise grid	σ_{noise}	0.1–3.5 MHz, step 0.001	—
Samples per σ_{noise}	—	60	—
Smoothing (mean / var)	—	EWMA 50 / 80	—

^a 4000 for the curriculum runtime sweep of Sec. VII.

^b Up to 150 per target in the curriculum sweep.

^c 0.40 in the curriculum sweep, which also uses it as the advancement threshold.

TABLE II. Architecture sweep on $N(2.2, 2.2, \pi/2)$, averaged over three seeds. Best value in each column in bold.

Hidden sizes	$\bar{C}_{\text{eval}}$	$\bar{F}_{\text{eval}}$	$\bar{F}_{\sigma=1}$	$\bar{T}$ (ns)	$\bar{L}$ (10^{-4})
(64, 64, 64)	0.3584	0.9941	0.9637	106.7	4.05
(64, 32, 32)	0.3592	0.9950	0.9622	111.3	3.67
(64, 64, 64, 32)	0.3626	0.9926	0.9604	107.3	4.17
(128, 64, 32)	0.3630	0.9920	0.9616	106.7	4.22
(128, 64, 32, 32)	0.3691	0.9906	0.9589	110.7	3.56

Two conclusions follow. First, architecture is not neutral: moderate changes in width and depth produce measurable differences in the final trade-off. Second, the dependence is not monotonic. The deepest model tested achieves the *lowest* mean leakage of the sweep and simultaneously the *worst* aggregate cost, the weakest robustness and the lowest nominal fidelity (Fig. 3). Minimizing one term of a multi-objective cost is not a selection principle.

The best aggregate performer is (64, 64, 64), but its

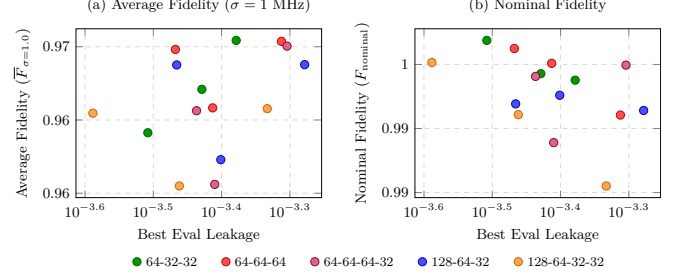


FIG. 3. Architecture trade-offs: evaluation leakage against (a) average fidelity at $\sigma_{\text{noise}} = 1$ MHz and (b) nominal fidelity. Leakage minimization is not a proxy for robustness — the layout with the lowest leakage, (128, 64, 32, 32), has the worst aggregate cost and the weakest robustness.

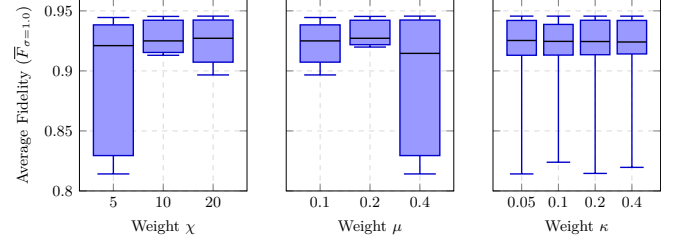


FIG. 4. Sensitivity of the average fidelity at $\sigma_{\text{noise}} = 1$ MHz to the UFO weights, shown separately for the fidelity weight χ , the boundary weight μ and the runtime weight κ . Underweighting fidelity ($\chi = 5$) opens a failure mode near $\bar{F} \approx 0.82$; $\mu = 0.4$ is strongly bimodal; κ has almost no effect in this fixed-horizon regime.

margin over the paper layout (64, 32, 32) is 0.2% in evaluation cost, and (64, 32, 32) is in fact better in mean nominal fidelity and mean leakage. Given a margin that small, and a sweep run under a deliberately reduced budget — which makes this a comparison of practical trainability rather than of asymptotic capacity — we retain the paper architecture (64, 32, 32) for all production runs. This maximizes comparability with Ref. [4] at negligible empirical cost.

B. Sensitivity to the UFO weights

The weights in Eq. (8) are part of the physical definition of the problem, not optimizer settings, so they deserve separate treatment. We swept $\chi \in \{5, 10, 20\}$, $\mu \in \{0.1, 0.2, 0.4\}$, $\kappa \in \{0.05, 0.1, 0.2, 0.4\}$ at fixed $\beta = 10$ (36 configurations, two seeds each). Holding β fixed makes the sweep a study of the ratios χ/β , μ/β , κ/β , which is the physically meaningful reading and keeps the design tractable.

A methodological caveat governs the whole sweep: the best evaluation cost is *not* a valid selection criterion across configurations, because different weights define different scalarizations and their numerical costs are not commensurable. Since all runs here terminate at the

same evaluation time of 180 ns, increasing κ mostly adds a rigid offset. We therefore compare on physical quantities only — nominal fidelity, leakage, and robustness at $\sigma_{\text{noise}} = 1$ MHz.

The fidelity weight has the clearest effect (Fig. 4). At $\chi = 5$ the distribution is broad and several runs collapse to $\bar{F} \approx 0.81$ –0.83; at $\chi = 10$ and $\chi = 20$ it tightens around 0.92–0.95. Underweighting logical accuracy relative to leakage lets the optimizer settle for controls that are clean but simply not the right gate. The boundary weight behaves non-monotonically: $\mu = 0.2$ gives the best median trade-off, $\mu = 0.1$ leaves slightly more leakage, and $\mu = 0.4$ becomes strongly bimodal, producing both some of the best runs of the sweep and its worst lower-tail failures. Larger μ does reduce the leakage proxy on average — the lowest-leakage region is often at $\mu = 0.4$ — but this does not translate into better robust control, which is the same lesson the architecture sweep taught. The runtime weight κ is nearly inert here, for the structural reason noted above; its influence reappears in Sec. VII, where gate duration is the observable rather than a fixed boundary condition.

The Pareto-favourable region is occupied jointly by $(\chi, \mu) \approx (10, 0.2)$ and $(\chi, \mu) \approx (20, 0.4)$, with the latter marginally ahead. Given two seeds and a small budget, that margin does not justify departing from the reference scalarization, so we retain

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1) \quad (17)$$

throughout. The useful conclusion is not the choice itself but its two components: the weights matter, in a physically interpretable way, and the original values sit close to the best region in an independent implementation.

V. SINGLE-TARGET BENCHMARK

All controllers are first compared on

$$U_{\text{target}} = N(2.2, 2.2, \pi/2), \quad (18)$$

a target that is neither near-identity nor a degenerate special case, that sits in the family used for the original robustness analysis, and that isolates optimizer differences by holding target difficulty fixed.

A. Direct gradient baseline

Under deterministic re-evaluation the best Adam solution is at $T^* = 70$ ns, with nominal cost 0.1576, fidelity 0.99662, total leakage 2.82×10^{-4} and boundary penalty 4.28×10^{-3} . Direct pulse optimization thus reaches fidelity comfortably above 99% with leakage in the 10^{-4} regime — exactly the scale the leakage-aware objective is designed to produce.

The horizon dependence (Fig. 5) is more informative than the optimum itself. Very short pulses fail badly:

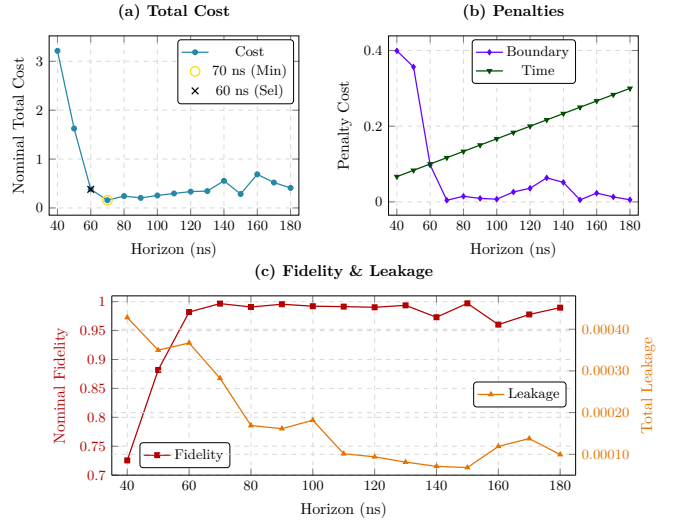


FIG. 5. Adam baseline on $N(2.2, 2.2, \pi/2)$ versus gate horizon. (a) Nominal total cost, with the optimum at 70 ns and the plan selected by the noisy objective at 60 ns both marked. (b) Boundary and runtime penalties. (c) Nominal fidelity and total leakage. The optimum arises from the simultaneous collapse of the boundary term and the rise in fidelity; longer horizons saturate in fidelity and are penalized by runtime.

fidelity drops to 0.726 at 40 ns and 0.882 at 50 ns, while the boundary penalties stay very large (0.399 and 0.357). Notably, the leakage remains in the same 10^{-4} band throughout this regime. The failure mode of ultra-short solutions is therefore *not* a leakage blow-up but the impossibility of satisfying accuracy and boundary regularity simultaneously under strong time compression — the speed-smoothness tension implicit in Eq. (6). This is visible sharply between 60 and 70 ns: fidelity rises from 0.982 to 0.997, the boundary term collapses from 9.75×10^{-2} to 4.28×10^{-3} , total cost falls from 0.381 to 0.158, while leakage improves only moderately.

The long-horizon side makes the multi-objective structure equally concrete. At 150 ns the fidelity is 0.99708 — slightly *better* than at the optimum — and the leakage is lower still, 6.82×10^{-5} ; yet the total cost rises to 0.285. Judged on fidelity alone the longer pulse would look preferable. Judged on the full objective it is not. The best solution is neither the most accurate nor the least leaky, but the best-balanced one.

Compared against the compiled reference of Eq. (12), the 70 ns Adam solution is already faster by $215/70 \approx 3.1$, before any policy learning is introduced. Analog optimization on this hardware model exploits structure that a fixed compiled implementation cannot.

B. Nominal TRPO

Trained in the deterministic environment with the same model and weights, the best TRPO policy reaches nominal cost 0.2982, fidelity 0.99415, leakage 1.60×10^{-3}

TABLE III. Nominal performance on $N(2.2, 2.2, \pi/2)$; best value in each column in bold. Two Adam rows are given: the plan selected by deterministic re-evaluation (70 ns) and the plan selected by the noisy training objective (60 ns), the latter being the one carried into the robustness study of Sec. VI. The gate-synthesis row is the compiled reference of Eq. (12) [4], shown for runtime only.

Controller	T (ns)	C^{nom}	F^{nom}	L_{tot}
Adam, best T	70	0.1576	0.99662	2.82×10^{-4}
Adam, robust plan	60	0.3806	0.98205	3.67×10^{-4}
TRPO, nominal	116	0.2982	0.99415	1.60×10^{-3}
TRPO, noise-opt.	90	0.1840	0.99876	2.12×10^{-3}
Gate synthesis	215	—	—	—

and an effective runtime of 116 ns. Two features deserve comment. The agent stops well before the 180 ns ceiling, confirming that it genuinely co-optimizes shape and duration rather than filling the available window. And 116 ns is already a factor $215/116 \approx 1.85$ below the synthesis reference, so even without noise-aware training the analog approach retains a runtime advantage — though a far more modest one than the order-of-magnitude gains reported for the best hardware-aligned targets.

C. Noise-optimized TRPO and nominal comparison

The noise-optimized agent, trained under $\sigma_{\text{train}} = 1$ MHz, reaches nominal cost 0.1840, fidelity 0.99876, leakage 2.12×10^{-3} and runtime 90 ns. Figure 6 and Table III collect all four solutions.

The comparison that matters for the rest of the paper is the third row against the first. On this deterministic benchmark the direct optimizer beats the policy-gradient controller simultaneously in cost (1.89 $\times$), runtime (1.66 $\times$), leakage (5.67 $\times$) and, marginally, fidelity (2.5×10^{-3} absolute). This is not a defect of the RL formulation; it is a delimitation of what it should be expected to deliver. When the model is known exactly and the objective is smooth and differentiable, direct gradient methods are extremely effective on fixed-target closed-system pulse design [3, 14], and nothing about policy optimization changes that. It also means the baseline against which the robustness claims will be tested is a strong one rather than a strawman.

The noise-optimized agent is the interesting case. It attains the *highest* nominal fidelity of all four solutions while also being the controller trained under perturbations — so whatever robustness it acquires is not bought by sacrificing nominal quality. Its leakage is the largest of the set, which is consistent with a policy that hedges against amplitude fluctuations rather than driving the multilevel dynamics as tightly as possible.

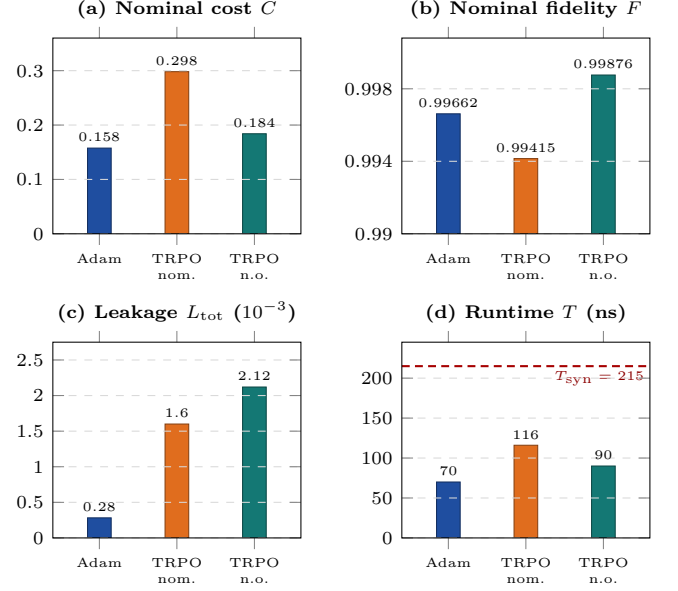


FIG. 6. Nominal comparison on $N(2.2, 2.2, \pi/2)$ of the Adam baseline at its best horizon, the nominal TRPO controller and the noise-optimized TRPO controller; values are those of Table III. (a) Total cost. (b) Nominal fidelity. (c) Total leakage. (d) Effective runtime, with the gate-synthesis reference $T_{\text{syn}} = 215$ ns marked. Both TRPO controllers produce genuinely high-quality solutions, but the direct optimizer dominates the nominal agent on all four deterministic metrics, and every analog solution is well inside T_{syn} .

VI. ROBUSTNESS UNDER STOCHASTIC CONTROL NOISE

A. Evaluation protocol

Each stored control plan is executed under the noise model of Sec. III B on a dense grid of 3401 noise strengths spanning $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz in steps of 0.001 MHz, with 60 independent realizations at each point — about 2×10^5 noisy rollouts per controller. Sixty samples give a noisy point estimate, so the curves are read through an exponentially weighted moving average along the σ_{noise} axis (span 50 for the mean, 80 for the variance). This is not cosmetic smoothing of sparse data: because neighbouring σ_{noise} values differ by 0.001 MHz, the average pools thousands of statistically comparable realizations, and the underlying scatter is retained in the figures. Beyond this smoothing, no post-processing of any kind is applied: every curve below is the direct Monte Carlo output of the evaluation pipeline.

An important detail of the comparison is that the Adam plan carried into this section is the one selected under the *noisy* training objective (Table III), so the baseline is itself noise-aware. The comparison is thus between a noise-aware direct optimizer and a noise-trained policy, not between a robust method and a nominal one.

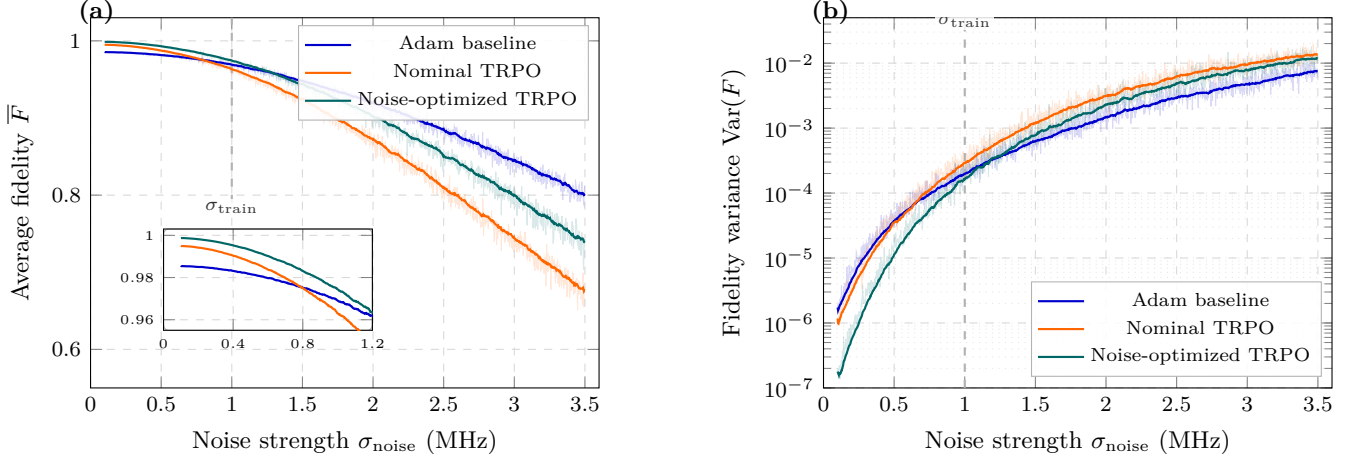


FIG. 7. Robustness on $N(2.2, 2.2, \pi/2)$. (a) Average fidelity $\bar{F}(\sigma_{\text{noise}})$ over $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz, with the low-noise regime magnified in the inset. (b) Fidelity variance on a logarithmic axis. Solid lines are exponentially weighted moving averages along the noise axis (spans 50 and 80); the underlying Monte Carlo scatter is visible behind them. The dashed vertical line marks the training noise $\sigma_{\text{train}} = 1$ MHz. The noise-optimized agent is above the nominal agent in (a) and below it in (b) at every noise strength, but it crosses the noise-aware Adam baseline at $\sigma_{\text{noise}} \approx 1.3$ MHz in the mean and ≈ 1.2 MHz in the variance, beyond which the direct optimizer is both more accurate and more stable.

B. Average fidelity and variance

All three controllers degrade monotonically with noise, as they must; the question is how fast, and relative to whom. Figure 7 separates two comparisons that the original framing tends to merge, and they give different answers.

The first is the controlled comparison between the two TRPO agents, which differ *only* in the distribution of trajectories seen during training. Here the result is unambiguous and holds without exception: the noise-trained policy has higher average fidelity and lower fidelity variance than the deterministically trained policy at every one of the 3401 noise strengths evaluated. Whatever is responsible cannot be reinforcement learning as such, since both agents use it; it can only be stochastic training.

The second is the comparison against the noise-aware Adam baseline, and here the picture is qualitatively different. The noise-trained policy leads only at low noise. The mean-fidelity curves cross at

$$\sigma_{\text{noise}}^{\times} \approx 1.29 \text{ MHz}, \quad (19)$$

and the variance curves cross slightly earlier, at ≈ 1.19 MHz; beyond these points the direct optimizer is simultaneously more accurate and more stable, and the gap widens monotonically. The noise-trained policy is the best of the three controllers on only 35% of the tested range in mean fidelity and 32% in variance — namely the part at and below the training noise.

Table IV makes both effects quantitative.

Against the nominal agent the improvement is present everywhere but shrinks with noise: the infidelity ratio falls from 3.76 at 0.1 MHz to 1.44 at 1 MHz and 1.24

TABLE IV. Robustness at representative noise strengths for the Adam baseline, the nominal TRPO agent (nom.) and the noise-optimized TRPO agent (n.o.). $\bar{F}$ is the channel average fidelity, Eq. (15); the last two columns give infidelity ratios $(1 - \bar{F}_X)/(1 - \bar{F}_{\text{n.o.}})$ in the upper block and variance ratios $\text{Var}(F)_X/\text{Var}(F)_{\text{n.o.}}$ in the lower block. Noise strengths σ_{noise} are in MHz; all values are EWMA estimates over the dense noise grid. Ratios below unity mark the regime, above $\sigma_{\text{noise}} \approx 1.3$ MHz, in which the direct optimizer overtakes the noise-trained policy.

σ_{noise}	$\bar{F}$			infidelity ratio	
	Adam	nom.	n.o.	nom./n.o.	Adam/n.o.
0.1	0.9857	0.9950	0.9987	3.76	10.88
0.5	0.9816	0.9874	0.9930	1.81	2.64
1.0	0.9691	0.9631	0.9744	1.44	1.21
1.5	0.9480	0.9238	0.9439	1.36	0.93
2.0	0.9193	0.8722	0.9020	1.31	0.82
3.0	0.8453	0.7449	0.7999	1.28	0.77
3.5	0.7989	0.6764	0.7381	1.24	0.77
Fidelity variance $\text{Var}(F)$ and ratios					
0.1	1.7×10^{-6}	1.2×10^{-6}	1.8×10^{-7}	6.56	9.67
1.0	2.0×10^{-4}	2.9×10^{-4}	1.7×10^{-4}	1.76	1.20
2.0	1.5×10^{-3}	3.1×10^{-3}	2.3×10^{-3}	1.37	0.65
3.5	7.6×10^{-3}	1.4×10^{-2}	1.2×10^{-2}	1.14	0.64

at 3.5 MHz, and the variance ratio from 6.56 to 1.76 to 1.14. Against the Adam baseline the ratios cross unity between 1 and 1.5 MHz and then invert: at 3.5 MHz the direct optimizer has 0.77 times the infidelity and 0.64 times the variance of the noise-trained policy.

The largest ratios sit at the lowest noise, and they are the least informative ones. At $\sigma_{\text{noise}} = 0.1$ MHz the comparison is dominated by the different *nominal* operating points of the three stored plans (Table III) rather than

by differential robustness. The experimentally relevant statement is the one at $\sigma_{\text{noise}} \approx 1$ MHz, where the noise-trained policy leads the nominal agent by $1.44\times$ in infidelity — a real and systematic improvement, but one to two orders of magnitude smaller than the gains reported in Ref. [4].

A nominal-independent summary is how much fidelity each controller loses across the whole sweep, $\Delta\bar{F} = \bar{F}(0.1 \text{ MHz}) - \bar{F}(3.5 \text{ MHz})$:

$$\begin{aligned} \Delta\bar{F}_{\text{Adam}} &= 0.187 < \Delta\bar{F}_{\text{noise-opt.}} = 0.261 \\ &< \Delta\bar{F}_{\text{nominal}} = 0.319. \end{aligned} \quad (20)$$

Stochastic training reduces the loss of the RL controller by 18% relative to deterministic training. It does not, however, make it the slowest-degrading controller: that is the noise-aware direct optimizer, by a clear margin.

C. Interpretation

Three conclusions follow, and they are more sharply delimited than a single benchmark number would suggest.

First, robustness is genuinely learned rather than inherited. The two TRPO agents share architecture, objective and algorithm, and differ only in the environment they were trained in; the noise-trained one is better on both metrics at every noise strength. The improvement is also two-dimensional — mean fidelity is not bought at the cost of shot-to-shot stability — which is what one expects if the policy has converged onto trajectories intrinsically less sensitive to amplitude fluctuations and to the unactuated anharmonicity parameter.

Second, and this is the main qualification our data impose, the learned robustness is *local to the training scale*. The agent was trained at $\sigma_{\text{train}} = 1$ MHz, and Eq. (19) places its advantage over a noise-aware direct optimizer at roughly $1.3\times$ that value. Below the crossover the policy exploits structure specific to the perturbation magnitude it was exposed to; above it, that specialization stops paying, and a controller that simply optimizes the same objective directly extrapolates better. This is a natural failure mode for distributionally trained policies, and it is invisible if robustness is quoted at a single operating point — which is precisely how it is usually quoted.

Third, the practical regime in which the learning formulation earns its cost is therefore narrow but well defined. It is the regime where the perturbation model is known well enough to be simulated *and* the deployed noise is close to the modelled noise. Where the noise level is uncertain, or where it may exceed the training scale by more than about 30%, a strong noise-aware direct optimizer is the safer choice on this problem. This does not contradict the mechanism the UFO framework proposes; it bounds the conditions under which the mechanism is worth its computational cost.

VII. RUNTIME ACROSS THE GATE FAMILY

A. Curriculum protocol

To test hardware efficiency beyond a single target we sweep the one-parameter subfamily

$$U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2), \quad \alpha \in [0, \pi], \quad (21)$$

in 21 steps of 0.157. Rather than retraining from scratch at each point, the best checkpoint obtained at one α initializes training at the next — a curriculum that exploits continuity in target space, in the spirit of the transfer strategy of Ref. [4]. Training at each target stops when the cost falls below 0.40 or after 150 iterations. The environment remains stochastic at $\sigma_{\text{train}} = 1$ MHz, so the runtimes reported here belong to robustly trained controllers, not to nominal ones.

As a no-transfer control we also sweep the same family with the Adam baseline, optimizing each of 33 targets independently over the horizon grid of Table I. This separates two effects that the curriculum conflates: how uniformly hard the analog control problem is across the family, and how much is gained by reusing previously learned structure.

B. Runtime landscape

The runtime profile is strongly structured rather than monotonic (Fig. 8, Table V). Four regimes are visible: a cold-start point, an irregular low- α region, a broad central basin of fast and accurate solutions, and a degradation region approaching $\alpha = \pi$.

The point $\alpha = 0$ must be read separately. It is the only target not warm-started from a neighbour, and it behaves like it: fidelity 0.558, runtime saturated at 180 ns, leakage 4.8×10^{-3} , robust fidelity 0.613. It measures cold-start difficulty, not family structure.

From $\alpha \approx 0.157$ the curriculum enters its transferable regime, reaching 142 and 140 ns with fidelities above 0.988. The interval $\alpha \in [0.47, 0.79]$ is then harder again — runtime returns to the 180 ns ceiling, fidelity falls to 0.948–0.954 and robust fidelity to 0.906–0.918 — and, tellingly, this region also shows visibly larger leakage and boundary costs. The difficulty is dynamical, not merely temporal.

The central basin $\alpha \in [1.1, 2.4]$ is the main result. Targets are solved in 76–140 ns, with the best stretch between $\alpha \approx 1.88$ and 2.36 giving 108, 106, 110 and 118 ns at nominal fidelity above 0.982 and robust fidelity 0.954–0.959. The speedup is not bought with accuracy: the fast region is also the accurate and robust region.

At the centre, $\alpha = \pi/2$ is a singular point. The controller terminates after 2 ns with fidelity 0.999999, leakage 4×10^{-10} , and robust fidelity still 0.9994 at $\sigma_{\text{noise}} = 1$ MHz — and it needs a single curriculum iteration to get there. This is not a fluctuation; it is a dynamical alignment, discussed below.

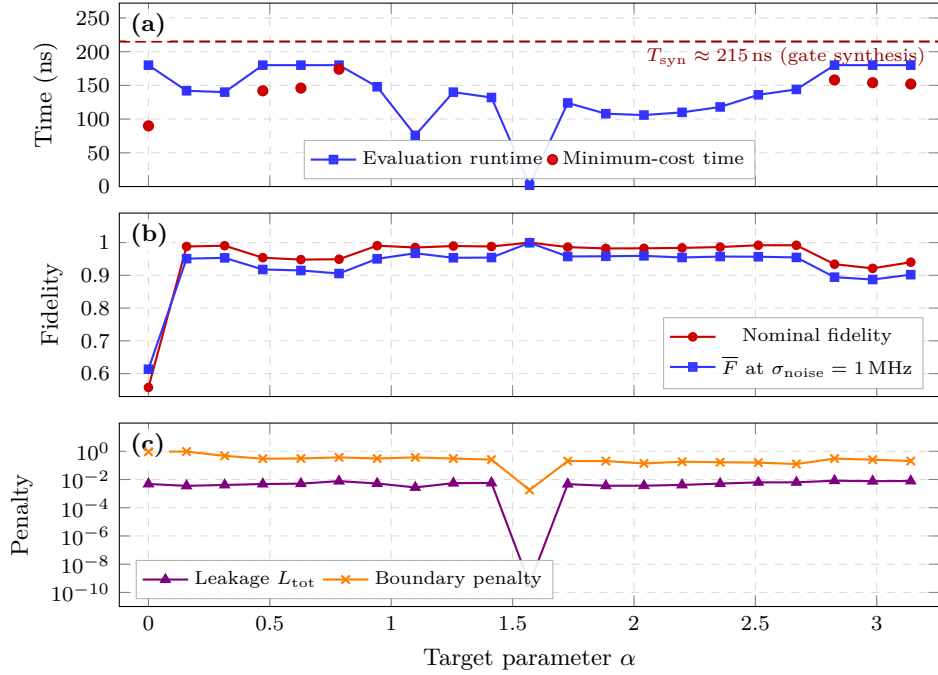


FIG. 8. Curriculum sweep over $N(\alpha, \alpha, \pi/2)$. (a) Final evaluation runtime and the minimum-cost time reached within the same training phase; the two separate exactly where the optimizer fails to push the cost below the early-termination threshold and the rollout runs to the 180 ns ceiling. The dashed line is the optimal gate-synthesis reference $T_{\text{syn}} \approx 215$ ns, which the analog controller stays below across the entire family. (b) Nominal fidelity and average fidelity under $\sigma_{\text{noise}} = 1$ MHz. (c) Leakage and boundary penalties on a logarithmic axis. The fastest regions are also the cleanest and the most robust.

Beyond the basin, $\alpha \in [1.73, 2.67]$ remains favourable (124–144 ns, fidelity 0.982–0.992), and only for $\alpha \gtrsim 2.83$ does the sweep degrade clearly, with runtimes back at 180 ns and fidelity falling to 0.921–0.940.

A diagnostic hidden in Fig. 8(a) is worth extracting. For well-converged targets the final runtime coincides with the minimum-cost time reached during the phase. In the hard regions it does not: near $\alpha \approx 0.47$ and 0.63 the lowest-cost trajectory occurs at 142–146 ns while the final rollout still ends at 180 ns, and the same happens near $\alpha \gtrsim 2.83$ with minima at 152–158 ns. The hard regions are therefore not regions where short solutions fail to exist — the agent visits them transiently — but regions where short solutions fail to be *stabilized* under the full objective.

C. Comparison with gate synthesis

Measured against $T_{\text{syn}} \approx 215$ ns, the analog controller is faster *everywhere* in the family. Even the hardest edge points, where the controller saturates the 180 ns ceiling, retain a 35 ns margin, i.e. about 16%. That the advantage survives precisely where the optimization is most difficult is the more robust half of the claim.

In the central basin the margin is systematic rather than marginal: for $\alpha \in [1.73, 2.36]$ the learned runtimes of 106–124 ns correspond to reductions of 42–51%, at

nominal fidelity 0.982–0.986 and robust fidelity 0.955–0.959. At $\alpha \approx 1.10$ the runtime drops to 76 ns, a factor 2.8, still above 0.98 nominal and 0.96 robust fidelity. And at $\alpha = \pi/2$ the 2 ns solution formally corresponds to a factor of order 100 — a number that should not be read as a family-level statement, but which does demonstrate something conceptually sharp: the compilation overhead of discrete synthesis is not a lower bound on physical execution time, only on execution time *through a prescribed gate alphabet*.

That distinction deserves emphasis, because it is easy to overclaim here. The synthesis benchmark is optimal within a circuit model over a fixed elementary gate library, whereas the analog controller optimizes in the continuous Hamiltonian-control space. The two solve related but different problems. Nothing in these results says that a quantum speed limit has been beaten. What they do say is that when the hardware offers a rich native interaction and sufficiently flexible pulse shaping, the shortest useful implementation of a target gate need not coincide with its shortest decomposition into universal primitives — which is the notion of efficiency that matters for near-term devices.

TABLE V. Curriculum runtime sweep over $N(\alpha, \alpha, \pi/2)$. T is the final evaluation runtime, $T_{\min}$ the minimum-cost time within the same phase, and the speedup is T_{syn}/T with $T_{\text{syn}} = 215$ ns. Rows in the central basin are the fastest.

α	T (ns)	$T_{\min}$ (ns)	F^{nom}	$\bar{F}_{\sigma=1}$	T_{syn}/T
0.000	180	90	0.5577	0.6133	1.19
0.157	142	142	0.9880	0.9510	1.51
0.314	140	140	0.9905	0.9533	1.54
0.471	180	142	0.9538	0.9179	1.19
0.628	180	146	0.9480	0.9149	1.19
0.785	180	174	0.9492	0.9055	1.19
0.942	148	148	0.9905	0.9504	1.45
1.099	76	76	0.9850	0.9674	2.83
1.256	140	140	0.9894	0.9535	1.54
1.413	132	132	0.9880	0.9542	1.63
1.571	2	2	0.99999	0.9994	107.5
1.727	124	124	0.9861	0.9575	1.73
1.884	108	108	0.9821	0.9582	1.99
2.041	106	106	0.9825	0.9595	2.03
2.198	110	110	0.9837	0.9545	1.95
2.355	118	118	0.9863	0.9573	1.82
2.512	136	136	0.9919	0.9567	1.58
2.669	144	144	0.9918	0.9547	1.49
2.826	180	158	0.9340	0.8944	1.19
2.983	180	154	0.9213	0.8872	1.19
3.140	180	152	0.9400	0.9020	1.19

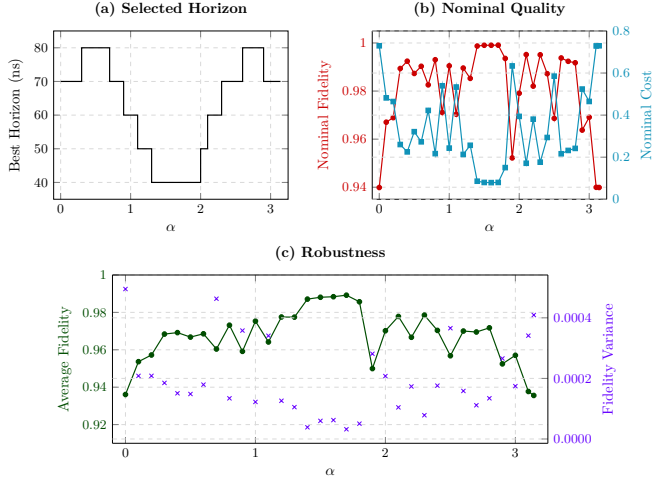


FIG. 9. No-transfer control: the Adam baseline swept independently over 33 targets of $N(\alpha, \alpha, \pi/2)$. (a) Selected horizon versus α . (b) Nominal fidelity and cost. (c) Average fidelity at $\sigma_{\text{noise}} = 1$ MHz and the corresponding variance. Difficulty is markedly non-uniform across the family even without any policy learning, with the easiest region in the centre — the same region the curriculum finds easiest.

D. What the runtime comparison actually measures

Two observations complicate the reading of Sec. VII and should be stated explicitly.

First, the independently optimized Adam baseline is

faster than the curriculum controller across the family. Its selected horizons span 40–80 ns with a median of 70 ns (Fig. 9), against 76–180 ns and a median near 140 ns for the curriculum TRPO sweep, at comparable nominal fidelity (median 0.987 versus 0.985) and comparable or better robustness at $\sigma_{\text{noise}} = 1$ MHz (median 0.969 versus 0.953). Every one of the 33 Adam solutions lies below T_{syn} , with a median speedup of $3.1\times$ against the $1.5\text{--}2.0\times$ typical of the curriculum. The single point at which the RL controller is decisively better is the singular $\alpha = \pi/2$.

The conclusion we draw is therefore narrower than a straightforward reading of Fig. 8 would support: the runtime advantage over compiled gate synthesis is a property of *direct analog optimization on this hardware model*, not a benefit conferred by reinforcement learning. Both optimizers exploit the native exchange interaction; the alignment argument of Sec. VIII explains the shape of both curves, and the family-resolved structure of Fig. 9 closely tracks that of Fig. 8.

Second, the curriculum runtimes are partly an artifact of the stopping rule. Training at each target halts as soon as the UFO cost drops below the advancement threshold 0.40, and the evaluation costs in Table V show the consequence directly: every converged phase sits in the narrow band $C \in [0.393, 0.400]$, immediately beneath the threshold, while the failures sit above 0.94. The agent therefore stops improving the moment it becomes good enough to advance, and has no incentive to shorten the pulse further. The curriculum runtimes should be read as *times sufficient to reach a fixed cost target* under a bounded budget, not as the shortest times the policy could achieve. A tighter threshold would very likely move the curve down; we did not run that sweep, and we flag it as the obvious next test rather than assuming its outcome.

E. Physical interpretation

The structure of Table V is not an artifact of the optimizer. The native entangling term of Eq. (1) is the exchange coupling $g(t)(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2)$, whose action on the computational subspace is aligned with the $XX + YY$ sector — exactly the sector that carries the α dependence of the target family, Eq. (11). When the required nonlocal content can be generated largely along that native direction, the controller does not have to fight the hardware: it can accumulate the entangling action directly and use the local channels only to shape phases, suppress leakage and satisfy the boundary constraints. Short runtimes in that regime are physically expected, not fortuitous.

This explains why the favourable region sits in the middle of the interval rather than at its edges, and why runtime, leakage and robustness co-vary across the sweep. In the basin the pulse needs only modest corrective excursions in the local channels, the boundary term stays moderate and the leakage proxy stays controlled; the re-

gion that is fastest is also the one that is easiest to control reliably. Toward the edges the target is less directly compatible with the native evolution, the controller must lean harder on detuning and microwave components to steer the dynamics while suppressing population transfer, and the consequence is twofold: longer sequences, because the correct effective nonlocal action takes longer to accumulate without violating the penalties; and more fragile ones, because a longer, more structured pulse is exposed to perturbations for longer. Both effects are visible in Fig. 8.

The $\alpha = \pi/2$ point should be read in the same spirit but with more caution. Its collapse to 2ns indicates a particularly favourable local-equivalence configuration within the family, in which the desired transformation is reached almost immediately once the controller is placed in the right dynamical sector. It is a statement about that point, not a generic realizability claim for two-qubit gates. The transferable conclusion is the pattern rather than the extremum: runtime minima occur where physical compatibility with the native generator is highest.

VIII. DISCUSSION

A. What reproduces

Two of the central claims of Ref. [4] reproduce cleanly under an independent implementation.

The first is that robustness against stochastic control errors is a *trainable* property rather than a post-hoc diagnostic. Our noise-optimized agent beats the otherwise identical nominal agent in both average fidelity and fidelity variance at every noise strength in $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz, and loses 18% less fidelity across the sweep, Eq. (20). Because the two agents differ only in the training distribution, the attribution is unambiguous. This is the claim of Ref. [4] that survives most cleanly.

The second is the physical mechanism behind the runtime advantage. Analog control is faster than optimal gate synthesis over the entire family, and — as predicted by the alignment argument — the gains are largest exactly where the target is closest to the native exchange interaction, with a singular point at $\alpha = \pi/2$. Both optimizers reproduce this structure, which supports not only the trend but the explanation offered for it.

B. What does not

Three quantitative discrepancies are worth stating plainly.

Nominal superiority does not hold. On $N(2.2, 2.2, \pi/2)$ the direct Adam baseline outperforms the nominal TRPO controller on every deterministic metric (Table III). Ref. [4] compares against an SGD baseline and reports a large margin in favour of policy optimization; with a well-tuned Adam optimizer acting on the same

Hamiltonian and the same objective, the margin reverses. We take this to mean that part of the originally reported gap reflects the strength of the baseline rather than an intrinsic property of policy-gradient control.

The robustness margin is much smaller, and it does not extend past the training scale. Against $10\times$ (versus noise-free RL) and $100\times$ (versus a gradient baseline) improvements in average gate infidelity, we find $1.44\times$ and $1.21\times$ at $\sigma_{\text{noise}} = 1$ MHz, the scale the original work itself identifies as experimentally representative (Table IV). Ratios approaching an order of magnitude appear only below $\sigma_{\text{noise}} \approx 0.2$ MHz, where they largely reflect the different nominal operating points of the compared plans rather than differential robustness. More consequential than the magnitude is the crossover of Eq. (19): above $\sigma_{\text{noise}} \approx 1.3$ MHz the noise-aware Adam baseline overtakes the noise-trained policy in *both* metrics and stays ahead. The original comparison is drawn against a gradient baseline optimized without noise awareness and reported at a single noise level, a protocol that cannot expose this behaviour. We regard the locality of the learned robustness, rather than its size, as the most important qualification our reproduction adds.

The runtime advantage is non-uniform, and it is not attributable to reinforcement learning. Rather than a uniform factor-ten improvement, we find a structured landscape spanning $1.19\times$ at the edges to $2.83\times$ in the basin, plus one singular point; and regions exist where the controller transiently discovers short solutions and then fails to stabilize them. More importantly, the independently optimized direct baseline is faster than the curriculum controller across the family (median $3.1\times$ versus $1.5\text{--}2.0\times$ over T_{syn}) at comparable fidelity and robustness (Sec. VIID). The speedup over compiled synthesis is therefore evidence for analog pulse-level control, not for the policy-gradient method that produced one instance of it. Part of the gap is a stopping-rule artifact — the curriculum halts at a fixed cost threshold — but that qualification cuts against the RL runtime claim rather than for it.

C. Why the discrepancies are unsurprising

None of this invalidates the framework, and some of it is simply what independent reimplementations means. The two studies differ in discretization, filtering pipeline, stopping rules, curriculum schedule, training budget and evaluation protocol, and the underlying optimization problem is high-dimensional and non-convex; exact numerical coincidence is not the appropriate standard.

The parameter studies of Sec. IV support this reading concretely. Performance depends measurably but non-monotonically on the actor-critic architecture, and interpretably on the UFO weights: too small a fidelity weight degrades control quality outright, while too strong a boundary penalty reduces leakage at the cost of destabilizing the fidelity-robustness trade-off. That the original

choices sit close to the best region in an independent implementation is a nontrivial check in their favour. That the surrounding landscape is not flat is why quantitative claims from this class of experiment should be quoted with their configuration.

D. Limitations

The study is simulation-based. The model is physically motivated but truncated to three levels per mode under a rotating-wave description, and real devices add transfer-function nonlinearities, spurious modes, drift and measurement constraints that are not represented. The noise model is Gaussian and white, applied to control amplitudes and to η ; it does not cover quasi-static drift, temporally correlated noise, or open-system decoherence, so the robustness reported here should be read as robustness *to that model*. Whether it generalizes to qualitatively different perturbations is the single most valuable open question left by this work. Finally, the comparison is between one RL framework and one strong direct optimizer; it is not a ranking of control methodologies, and does not include automatic-differentiation GRAPE/Krotov variants, differentiable programming controllers, or model-based RL [11].

E. Relation to subsequent work

Read from the present vantage point, the lasting contribution of Ref. [4] looks less like a particular optimizer and more like a demonstration that robust analog control can be posed as a learning problem with leakage-aware costs and stochastic perturbations. The literature since has refined that in two directions that our results make concrete. Model-free and hardware-interacting schemes reduce model bias by adapting on the device itself [10]; and model-based or hybrid methods address the sample inefficiency that makes pure model-free RL expensive in practice [11]. Our results sharpen the case for both. A direct optimizer wins the nominal problem and wins again once the deployed noise exceeds the modelled noise; the learned policy wins in between. Since a real device’s noise is known only approximately, that is an argument for hybrid workflows rather than for either extreme: use direct optimization or a learned policy to generate physically informed pulses, then refine them in closed loop against the device, where the effective perturbation distribution is the device’s own rather than a modelled one [16–18].

IX. CONCLUSION

We have independently reimplemented and assessed the UFO framework for robust analog two-qubit gate design on a multilevel gmon model, including the second-

order TSWT leakage bound, the exact bandwidth filter, and TRPO training in a stochastic control environment.

The framework’s core mechanism survives independent scrutiny. Embedding the perturbation model in the training environment produces a controller that is measurably more accurate *and* more reproducible under control noise than the identical architecture trained deterministically, at every noise strength tested, losing 18% less fidelity across the sweep. The runtime advantage of analog control over optimal gate synthesis is likewise real across the whole family $N(\alpha, \alpha, \pi/2)$, and is largest exactly where the target aligns with the native exchange interaction — confirming the physical mechanism, not just the trend.

The reproduction also narrows the claims in three ways. Policy-gradient control is not a better nominal optimizer than a well-tuned direct gradient method on a fixed, exactly known target. The runtime advantage over compiled synthesis, while real and physically interpretable, belongs to analog pulse-level control rather than to the learning method: an independently optimized direct baseline is faster across the family, and only the singular point $\alpha = \pi/2$ favours the policy. And, most consequentially, the robustness advantage is local: measured against a direct optimizer given the same noise awareness, the noise-trained policy leads only up to $\sigma_{\text{noise}} \approx 1.3$ MHz, about 1.3 times the noise it was trained on, and falls behind in both mean fidelity and variance beyond that.

That last result is the one we would most like to see tested elsewhere. It suggests that the useful question about noise-aware RL for quantum control is not how large its advantage is at the design noise level, but how far in perturbation space that advantage extends — a quantity that a single-operating-point benchmark cannot reveal, and that determines whether such controllers are usable on a device whose noise is only approximately known. The natural companion experiment is the one our noise model does not reach: whether robustness trained on Gaussian amplitude fluctuations transfers at all to quasi-static drift, colored noise, parameter mismatch and decoherence. Together those two axes, magnitude and mechanism, separate a policy that has learned distribution-specific compensation from one that has found a genuine control regularity.

DATA AVAILABILITY

The robustness curves of Fig. 7 and Table IV are the direct Monte Carlo outputs of the evaluation pipeline on the grid of Table I, with no post-processing other than the exponentially weighted smoothing stated in Sec. VI; raw and smoothed columns are both retained in the released data. The runtime sweep of Table V and the model-selection sweeps of Sec. IV are likewise reported as produced by the training scripts.

The material needed to reproduce every number in

this paper consists of the simulation code implementing Secs. II and III, the stored control plans for the four controllers of Table III, the per-target logs of the curriculum and Adam family sweeps, and the per-seed records of the two model-selection sweeps. Together with the complete parameter set given in Table I, these are sufficient to regenerate all figures and tables from scratch. They are available from the author on request and are being prepared for public deposit.

ACKNOWLEDGMENTS

I thank Prof. Morten Hjorth-Jensen and Prof. Davide Emilio Galli for their supervision and for many discussions that shaped this work. This study was carried out as part of a Bachelor’s thesis at the Department of Physics, Università degli Studi di Milano. AI-based tools were used as auxiliary instruments for code drafting and debugging, literature search, and stylistic revision; all derivations, modeling assumptions, numerical methodology, experimental design and results were independently verified by the author.

Appendix A: The TSWT coherent leakage bound

We summarize the derivation of Eq. (6). Starting from the decomposition Eq. (4) in the regime $\epsilon/\Delta \ll 1$, one seeks an anti-Hermitian generator $\hat{S}(t) = \epsilon\hat{S}_1 + \epsilon^2\hat{S}_2 + \dots$ such that the transformed Hamiltonian Eq. (5) is block-diagonal to the desired order. Order by order, the block-off-diagonal components must satisfy

$$[\hat{H}_0, \hat{S}_1] = -\hat{H}_2, \quad [\hat{H}_0, \hat{S}_2] = -([\hat{H}_1, \hat{S}_1] - i\dot{\hat{S}}_1)_{\text{od}}, \quad (\text{A1})$$

which are solved elementwise on the off-diagonal blocks by dividing by the subspace energy differences $E_\alpha - E_{\alpha'}$; the assumption $\epsilon/\Delta \ll 1$ is what makes these divisions safe. After second order the residual block-off-diagonal Hamiltonian is

$$H_{\text{od}}^{(3)} = [\hat{H}_1, \hat{S}_2] + \frac{1}{3} [[\hat{H}_2, \hat{S}_1], \hat{S}_1] - i\dot{\hat{S}}_2 = O\left(\frac{\epsilon^3}{\Delta^2}\right). \quad (\text{A2})$$

The leakage amplitude out of Ω_0 over $[0, T]$ is then an interaction-picture integral of $\hat{H}_{\text{od}}$ against a rapidly oscillating phase $e^{i \int \Delta dt}$. Repeated integration by parts converts this into boundary terms, which pick up $\|\hat{H}_{\text{od}}\|/\Delta$ at $t = 0$ and $t = T$, plus a residual integral controlled by

the second derivative of $\hat{H}_{\text{od}}$ divided by Δ^2 . Collecting these gives Eq. (6).

The two contributions dominate in complementary regimes: for slow, off-resonant modulation the boundary terms dominate, and for faster on-resonant modulation the derivative term does. This is why the bound is usable as a single penalty across the operating range rather than only in one limit.

Appendix B: Algorithmic details

Environment step. Given the current observation and propagator: map the action to physical controls; apply the bandwidth filter Eq. (13); if stochastic training is enabled, sample Gaussian perturbations for $g, \delta_{1,2}, f_{1,2}$ and η and wrap the phases modulo 2π ; build H_t from Eq. (1); propagate $U_{t+1} = e^{-iH_t\Delta t}U_t$ by exact diagonalization; update the leakage estimator; evaluate the running cost Eq. (8) and reward $r = -C$; test the stopping condition (time limit or cost threshold); emit the next observation Eq. (14).

Incremental leakage estimator. At each step: decompose H_t into H_0, H_1, H_2 ; solve Eq. (A1) for S_1 ; estimate $\dot{S}_1$ from the stored previous value by finite differences; solve for S_2 ; estimate $\dot{S}_2$; form $H_{\text{od}}^{(3)}$ from Eq. (A2) and symmetrize; take its spectral norm. The first and last samples update the boundary contribution; the discrete second derivative, formed from the two previous stored values, updates the integral contribution. Storing only the two previous states keeps the estimator $O(1)$ per step.

TRPO loop. Per iteration: collect a batch of stochastic rollouts in parallel workers; compute discounted returns and generalized advantage estimates [19]; take the KL-constrained policy step Eq. (10) by conjugate gradient with backtracking line search; refit the value network by regression on the empirical returns; evaluate the deterministic policy in the nominal environment and store the control plan if it improves on the best so far. Checkpoints and induced control plans are saved every iteration, so that comparisons can be made at the level of the realized pulse rather than only at the level of network weights.

Adam baseline. For each candidate horizon: discretize the trajectory into $N_T = T/\Delta t$ steps; initialize the pulse parameters; iterate simulate–evaluate–update for 400 Adam steps; store the optimized plan and re-evaluate it deterministically for cost, fidelity, leakage, boundary penalty and runtime. The best horizon is selected on the re-evaluated metrics, not on the optimizer’s internal objective — a distinction that matters here, since the noisy objective selects 60 ns while deterministic re-evaluation selects 70 ns.

[1] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, *Applied Physics Reviews* **6**, 021318 (2019).

[2] M. Kjaergaard, M. E. Schwartz, J. Braumüller, P. Krantz, J. I.-J. Wang, S. Gustavsson, and W. D. Oliver, *Annual Review of Condensed Matter Physics* **11**,

- 369 (2020).
- [3] C. P. Koch, U. Boscain, T. Calarco, G. Dirr, S. Filipp, S. J. Glaser, R. Kosloff, S. Montangero, T. Schulte-Herbrüggen, D. Sugny, and F. K. Wilhelm, *EPJ Quantum Technology* **9**, 19 (2022).
 - [4] M. Y. Niu, S. Boixo, V. N. Smelyanskiy, and H. Neven, *npj Quantum Information* **5**, 33 (2019).
 - [5] Y. Chen, C. Neill, P. Roushan, N. Leung, M. Fang, R. Barends, J. Kelly, B. Campbell, Z. Chen, B. Chiaro, A. Dunsworth, E. Jeffrey, A. Megrant, J. Y. Mutus, P. J. J. O'Malley, C. M. Quintana, D. Sank, A. Vainsencher, J. Wenner, T. C. White, M. R. Geller, A. N. Cleland, and J. M. Martinis, *Physical Review Letters* **113**, 220502 (2014).
 - [6] F. Motzoi, J. M. Gambetta, P. Rebentrost, and F. K. Wilhelm, *Physical Review Letters* **103**, 110501 (2009).
 - [7] M. Werninghaus, D. J. Egger, F. Roy, S. Machnes, F. K. Wilhelm, and S. Filipp, *npj Quantum Information* **7**, 14 (2021).
 - [8] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, in *Proceedings of the 32nd International Conference on Machine Learning*, PMLR, Vol. 37 (2015) pp. 1889–1897.
 - [9] M. Bukov, A. G. R. Day, D. Sels, P. Weinberg, A. Polkovnikov, and P. Mehta, *Physical Review X* **8**, 031086 (2018).
 - [10] V. V. Sivak, A. Eickbusch, H. Liu, B. Royer, I. Tsioutsios, and M. H. Devoret, *Physical Review X* **12**, 011059 (2022).
 - [11] I. Khalid, C. A. Weidner, E. A. Jonckheere, S. G. Schirmer, and F. C. Langbein, *Physical Review Research* **5**, 043002 (2023).
 - [12] F. Vatan and C. Williams, *Physical Review A* **10.1103/PhysRevA.69.032315** (2004).
 - [13] D. P. Kingma and J. Ba, *International Conference on Learning Representations* (2015).
 - [14] N. Khaneja, T. Reiss, C. Kehlet, T. Schulte-Herbrüggen, and S. J. Glaser, *Journal of Magnetic Resonance* **172**, 296 (2005).
 - [15] M. A. Nielsen, *Physics Letters A* **303**, 249 (2002).
 - [16] P. J. Vetter, T. Reisser, M. G. Hirsch, T. Calarco, F. Motzoi, F. Jelezko, and M. M. Müller, *npj Quantum Information* **10**, 96 (2024).
 - [17] N. J. Glaser, F. A. Roy, I. Tsitsilin, L. Koch, N. Bruckmoser, J. Schirk, J. H. Romeiro, G. B. P. Huber, F. Wallner, M. Singh, G. Krylov, A. Marx, L. Södergren, C. M. F. Schneider, M. Werninghaus, and S. Filipp, *Physical Review Applied* **24**, 024048 (2025).
 - [18] Z. T. Wang, Q. Chen, Y. Du, Z. H. Yang, X. Cai, K. Huang, J. Zhang, K. Xu, J. Du, Y. Li, Y. Jiao, X. Wu, W. Liu, X. Lu, H. Xu, Y. Jin, R. Wang, H. Yu, and S. P. Zhao, *Quantum compiling with reinforcement learning on a superconducting processor* (2024), [arXiv:2406.12195 \[quant-ph\]](https://arxiv.org/abs/2406.12195).
 - [19] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, in *Proceedings of the International Conference on Learning Representations (ICLR)* (2016).